

MALEK HAMMOUD

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EDUCATION

McMaster University

Hamilton, ON

Honours Bachelor of Science in Computer Science (Co-op)

Sep 2026 – Apr 2030 (Expected)

- **Core Focus:** Data Structures & Algorithms, Systems Programming, Software Engineering Methodologies.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL (PostgreSQL), Bash, HTML/CSS

Full-Stack & Backend: React, Node.js, Express, REST APIs, Web Scraping Pipelines, PostgreSQL, Storybook

AI & Machine Learning: Computer Vision (YOLOv5, PyTorch), Multimodal LLM Integration, Scikit-learn, Pandas, NumPy

Systems, Cloud & Security: Linux (Arch/Debian), Embedded Systems (QNX/RTOS), Docker, Azure, Git, AppSec (RBAC/Auth), MAVLink/ArduPilot

EXPERIENCE

Playtoon IT, Inc.

Remote

Full-Stack Developer Intern

Mar 2025 – Aug 2025

- Shipped production features for an interactive web platform as part of a 15-member engineering team, taking work from spec through code review to release.
- Redesigned UI state workflows to cut redundant re-renders and streamline the feature-release feedback loop for the rest of the team.
- Built secure, automated media upload pipelines with real-time profanity filtering, removing manual content moderation from the release path.

Bryck Technologies

Remote

Software Developer Intern

Sep 2024 – Jan 2025

- Owned a full-stack internal enterprise web application end-to-end – from design through production deployment – becoming the team's primary point of contact for the system post-launch.
- Architected a reusable React component library in Storybook, cutting duplicate UI work across features and standardizing front-end patterns for the rest of the team.
- Managed PostgreSQL schema migrations inside an Agile workflow, keeping releases backward-compatible with zero downtime for existing internal users.

TECHNICAL PROJECTS

Shadow Stalker — 1st Place – Hack the North 2026

github.com/malekhammoud/Drone-Sim

Lead – Infra & Integration — Python, ArduPilot, MAVLink, Gazebo, PyTorch, Geodesy

Sep 2026

- Led infrastructure and integration for a 4-person team that won the Dominion Dynamics sponsor track and placed Top 32 of 250+ teams overall, building a fleet-wide search system that autonomously located a moving vessel across a live Gazebo/ArduPilot Arctic simulation.
- Built `arcticlib`, a shared telemetry/vision abstraction over MAVLink with one synchronized simulation clock, letting three teammates build detection, tracking, and planning in parallel with zero integration conflicts.
- Engineered pixel-to-GPS geolocation via camera-ray trigonometry and WGS84 geodesics, calibrating sensor mount angle to cut median localization error from 79m to 38m, then to 12.4m (2.2m best) after fixed-wing-to-quadcopter handoff.

Drone Litter Mapping & Recovery System — Published Research Paper malekhammoud.com/litter-detection

Creator & Lead Developer — ArduPilot, Pixhawk, YOLOv5, Postgres, React Native, ROS2

2025

- Built a full-stack outdoor autonomy pipeline: an ArduPilot/Pixhawk quadcopter flying GPS waypoint surveys, a YOLOv5 detector reaching **90% field accuracy**, and a Postgres-backed API rendering live GPS-tagged litter onto a React Native map.
- Achieved **~1m loiter/landing accuracy** at a 500g stable payload, handed detections to a separate ROS2 walking robot for recovery, and published the system as a peer-reviewable research paper.

IsMyTripSafe — Full-Stack / Automated Data Pipeline

ismytripsafe.com

Creator & Lead Developer — Python, Web Scraping, Programmatic SEO

2026

- Architected an autonomous destination safety intelligence platform serving **2,000+ monthly active users**.
- Engineered automated web scraping pipelines against international government datasets and travel advisory APIs, feeding dynamic report generation and a programmatic SEO layer that drives autonomous traffic growth.

Flow Arch (Linux OS) — SourceForge Rising Star Award

flowarch-os.github.io

Lead Developer & Maintainer — Arch Linux, Calamares, Shell Scripting

Dec 2025 – Present

- Engineered a custom Arch Linux distribution with **700+ downloads**; built using Arch ISO tools and Calamares installer.
- Awarded the **SourceForge Rising Star Award** for rapid open-source community adoption and distribution stability.

Basil Home AI (Smart Appliance Prototype) — Hack the 6ix Winner

basilhome.ca

Co-Creator & Embedded/AI Developer — QNX RTOS, YOLOv5, Multimodal LLMs

Jul 2025

- Engineered an AI smart appliance system on QNX RTOS/Raspberry Pi, training YOLOv5 vision models for real-time inventory tracking and integrating multimodal LLM pipelines for recipe generation.
- Awarded Deloitte's "Best Use of AI for Green" at Hack the 6ix for embedded resource efficiency and practical utility.

SECURITY RESEARCH

Independently run authorized Bugcrowd offensive-security engagements against AI agents/MCP servers and production APIs, applying VRT/CVSS 3.1 methodology in human-written disclosure reports.

Intercom (Fin AI): zero-click PII exfiltration via indirect prompt injection in markdown rendering – triaged and awarded.

LaunchDarkly: broken-access-control flaw exposing webhook signing secrets to low-privileged roles. **Recorded Future MCP:** prompt-injection chain reaching privileged write operations – active engagement.